METRIC

A-A-54656
15 May 1992

COMMERCIAL ITEM DESCRIPTION

IMMERSION OIL, MICROSCOPY

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-I-36698A which is cancelled.

This Commercial Item Description covers an immersion oil having 1,250 Centistoke viscosity, for use in microscopy.

Salient characteristics:

Shall be a nondrying immersion oil for microscopy.

Shall not contain any polychlorinated biphenyl (PCB) compounds.

Shall be a nondrying type oil which does not change its refractive index value when exposed to air. The immersion oil shall retain nondrying characteristics after standing in an uncovered vessel at room temperature for at least eight (8) hours, by feeling sticky to the touch of a finger without evidence of polymerization or setting.

Shall conform to the requirements of International Standard ISO 8036/1, except for the following:

The requirements for the optical properties shall be superseded by the requirements in this document.

The requirements for marking shall be superseded by the requirements in this document.

AMSC N/A FSC 6640

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Shall be formulated to give the following characteristics:

Standardized for use at 23°C ± 0.1°C, and at a pressure of 1 atmosphere.

A viscosity of 1,250 Centistokes ± 10% at 23°C, based on using an Ostwald-type viscometer or automated assembly for determining the viscosity of an immersion oil in accordance with testing methods in ASTM D445.

A refractive index of 1.5150 ± 0.0005 for D line of sodium light at 23°C, based on using a refractometer instrument (e.g. an Abbe' refractometer) for determining refractometer index.

An Abbe number of 44 ± 5 based on testing methods in ASTM D1807 and the Abbe number being calculated using the following formula:

$$\text{Abbe number} = \frac{nD-1}{nF-nC}$$

where

nD is the refractive index using 589.3 nm light

nF is the refractive index using 486.1 nm light

nC is the refractive index using 656.3 nm light

A Saybolt color of not less than +7, based on testing methods in ASTM D156.

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Shall be supplied in a light-resistant bottle. The bottle shall include a label that includes the following information:

The manufacturer's or supplier's name and location (i.e. city).

The principal refractive index and tolerances.

The Abbe number and tolerances or manufacturing specification.

The temperature of calibration.

The refractive index temperature coefficient.

The viscosity in centistokes and tolerance.

The lot number and date of expiration.

A statement that indicates that the immersion oil contains no polychlorinated biphenyl (PCB).

Workmanship. The immersion oil shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Bottle (BT). One bottle, containing one fluid ounce of immersion oil, as specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspection and Testing. Inspection and testing shall be conducted on the immersion oil in accordance with commercial practice. The sampling for examination and testing shall be in accordance with commercial practice.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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egulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Other publications. The following documents form a part of this specification to the extent that is specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposals shall apply.

ASTM D156	Standard Test Methods for Saybolt Color of Petroleum Products (Saybolt Chromometer Method).
ASTM D445	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity).
ASTM D1807	Standard Test Methods for Refractive Index and Specific Optical Dispersion of Electrical Insulating Liquids.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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ISO

ISO 8036/1 Optics and optical instruments - Microscopes Part 1: Immersion oil for general use in light microscopy.

(Application for copies should be addressed to American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

Ordering data. Expiration dating period required, intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

NSN	Item Identification
6640-00-299-9807	IMMERSION OIL, MICROSCOPY, 1 oz.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6640-1745

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

1 RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	A-A-54656	15 May 1992

3 DOCUMENT TITLE

IMMERSION OIL, MICROSCOPY

4 NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code)	
Defense Personnel Support Center	(1) Commercial (215) 737-2870	(2) AUTOVON 444-2870
c ADDRESS (Include Zip Code) ATTN: MSE 2800 S. 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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